

Module-1

Session-1

Q1. Name 5 IT job roles and write one line about what each role does.

Ans. **Software Developer** – Develops and maintains software applications.

Web Developer – Creates and maintains websites and web applications.

Software Tester – Tests software and finds bugs and errors.

Data Analyst – Analyzes data to find useful information.

AI Engineer – Develops AI and Machine Learning systems.

Q2. List 3 product companies and 3 service companies with one example of work each does.

Ans. **Product Companies:**

Google – Develops products like Google Search and Gmail.

Microsoft – Develops products like Windows and Office.

Apple – Develops products like iPhone and macOS.

Service Companies:

TCS – Provides IT consulting and software development services.

Infosys – Provides IT consulting and digital technology services.

Wipro – Provides IT and business services.

Q3. Write the full forms and one-line use of: GitHub, Slack, Jira, IDE, API.

Ans. **GitHub** – A platform used to store, manage, and collaborate on source code
Slack – A communication platform used for team communication and collaboration.

Jira – A project management tool used to track tasks, bugs, and projects.

IDE – Integrated Development Environment – Used to write, run, and debug programs.

API – Application Programming Interface – Allows different software applications to communicate with each other.

Q4. Name any 5 programming languages and mention one popular use case for each.

Ans. **Python** – Used for AI, Machine Learning, and Data Science.

Java – Used for enterprise and Android applications.

JavaScript – Used for websites and web applications.

C++ – Used for game development and high-performance software.

C# – Used for Windows applications and game development.

Q5. What is the difference between a developer and a tester? Answer in 2 lines.

Ans. A **Developer** creates software, develops new features, and fixes coding problems. A **Tester** checks the software, finds bugs, and ensures that it works correctly.

Session-2

Q1. What are the three main layers of software architecture? Write one line about each.

Ans. **Presentation Layer** – Shows the application to the user.

Business Logic Layer – Handles the main rules and processing of the application.

Data Layer – Stores and manages the application data.

Q2. Define client and server in your own words with one real-life example.

Ans. **Client** – A device or application that requests a service.

Server – A computer that provides the requested service or data.

Q3. Write 2 differences each between desktop apps, web apps, and mobile apps.

Ans. **Desktop app**- Runs on a computer.

Usually needs installation.

Web app- Runs in a web browser.

Usually does not need installation.

Mobile App- Runs on a smartphone/tablet

Usually installed from an app store.

Q4. Name any 3 databases and mention whether each is SQL or NoSQL.

Ans. **MySQL** – SQL

PostgreSQL – SQL

MongoDB – NoSQL

Session-3

Q1. Write the full form and one-line use of: HTTP, HTTPS, FTP, SMTP.

Ans. **HTTP**: hypertext transfer protocol

Used to transfer data between a web browser and web server.

HTTPS : hypertext transfer protocol secure

Used to transfer website data securely and in encrypted form.

FTP : file transfer protocol

Used to transfer files between computers or servers.

SMTP: simple mail transfer protocol used to send emails.

Q2. What is the difference between IPv4 and IPv6? Answer in 2 lines.

Ans. **IPv4**: Uses a 32-bit address and has a limited number of IP addresses.

IPv6: Uses a 128-bit address and provides a much larger number of IP addresses.

Q3. Run nslookup google.com on your system and write down the IP address you get.

Ans. The IP address of google.com can change depending on the system and location.

Example: 142.250.195.14

Q4. What do these HTTP status codes mean: 200, 301, 404, 500, 403? **200 – OK**: Request was successful.

Ans. **301** – Moved Permanently: Page has moved to a new URL.

404– Not Found: Page was not found.

500 – Internal Server Error: Server has a problem.

403 – Forbidden: Access is not allowed.

Q5. Explain DNS hierarchy (root, TLD, domain, subdomain) with `www.google.com` as an example

Ans. DNS converts a website name into an IP address.

Example: `www.google.com`

Root (.) -Top level of DNS.

TLD (.com) -Top-Level Domain.

Domain (google) -Main website name.

Subdomain (www) -Subdomain of google.com.

Order: Root → TLD → Domain → Subdomain

Session-4

Q1. List all 6 phases of SDLC in order with one line about each.

Ans. **Planning** – Decide what the software should do and make a plan.

Requirement Analysis – Collect and understand the user's requirements.

Design – Create the design and structure of the software.

Development – Write the code and build the software.

Testing – Check the software and find/fix errors.

Maintenance – Update the software and fix problems after release.

Q2. Write 3 differences between Agile and Waterfall models.

Ans. Agile :- Work is done in small parts.

Changes can be made easily

Testing is done regularly

Waterfall:- Work is done step-by-step

Changes are difficult to make

Testing is done after development

Q3. Name the 3 main Scrum roles and write one line about the responsibility of each.

Ans. Product Owner – Decides what the team should build and manages priorities.

Scrum Master – Helps the team follow Scrum and removes obstacles.

Developers – Build and deliver the product.

Q4. What is a Sprint in Scrum? Mention its typical duration.

Ans. A Sprint is a fixed period in which the Scrum team completes a set of tasks.

Q5. What do these Jira board columns mean: To Do, In Progress, Done, Backlog.

Ans. Backlog: Future tasks that need to be done.

To Do: Tasks ready to start.

In Progress: Tasks currently being worked on.

Session:-5

Q1. Write the Git command for each: initialize a repo, stage all files, commit with a message, push to remote, pull latest changes.

Ans. Initialize repository: `git init`

Stage all files: `git add .`

Commit with message: `git commit -m "message"`

Push to remote: `git push origin main`

Pull latest changes: `git pull origin main`

Q2. What is the difference between `git fetch` and `git pull`? Answer in 2 lines.

Ans. `git fetch` downloads new changes but does not merge them.

`git pull` downloads new changes and merges them into your branch.

Q3. Create a new GitHub repository, push any small text file from your local system, and share the repo link.

Ans. Create a new repository on GitHub and push a small text file using these commands:

`git init`

`git add .`

`git commit -m "Add text file"`

`git branch -M main`

`git remote add origin <repo-link>`

`git push -u origin main`

Repo Link: <https://github.com/your-username/your-repository>

Q4. What is the difference between a Fork and a Pull Request? Answer in 2 lines.

Ans. **Fork:** Creates a copy of someone else's repository in your GitHub account.

Pull Request: Requests the owner to review and merge your changes.

Q5. Create a new branch named feature-test, make one commit on it, and merge it back to the main branch. Write the commands you used.

Ans. `git checkout -b feature-test`

`git add .`

`git commit -m "Add feature"`

`git checkout main`

`git merge feature-test`

Create feature-test, make one commit, switch to main, and merge it.